Quantization Error Bounds and Embedding-Distance Invariance in Machine Learning

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Abstract—We extend the rescaling-invariance and fixed-bit-budget analysis of prior work to two concrete machine learning mechanisms: (1) uniform quantization of model weights/activations, where information loss is real and precisely bounded, and (2) cosine-similarity-based embedding search, where scaling is provably lossless. Both are proved exactly and verified against direct numerical simulation, continuing the same standard of evidence used throughout this series.

1 Uniform Quantization: Where Information Loss Is Real

Model compression often maps 32-bit floating-point weights to a low-bit fixed-point representation to reduce memory and increase inference speed. This is the machine-learning instance of the fixed-bit-budget scenario flagged in earlier work (Section IV.B of the prior paper): unlike a pure linear rescaling, quantization also *rounds* to a finite set of levels, and that rounding is where information is actually destroyed.

1.1 Model and bound

Let $x \in [x_{\min}, x_{\max}]$ be a weight value, and let a b-bit uniform quantizer partition this range into 2^b levels of width

$$\Delta = \frac{x_{\max} - x_{\min}}{2^b}. \tag{1}$$

Each x is mapped to the nearest level $\hat{x}$.

Proposition 1 (Maximum quantization error)**.** *For any $x \in [x_{\min}, x_{\max}]$, $|x - \hat{x}| \le \Delta/2$.*

Proof. By construction, $\hat{x}$ is the nearest of 2^b evenly spaced points covering an interval of width $x_{\max} - x_{\min}$, so consecutive levels are exactly Δ apart. Any x lies within $\Delta/2$ of its nearest level, since if it were farther than $\Delta/2$ from both surrounding levels, those levels would be more than Δ apart, a contradiction. □

Proposition 2 (Mean squared quantization error)**.** *Under the standard uniform-noise model (valid when Δ is small relative to the signal's variation, so rounding error is approximately uniformly distributed on $[-\Delta/2, \Delta/2]$),*

$$\mathbb{E}[(x - \hat{x})^2] = \frac{\Delta^2}{12}. \tag{2}$$

Proof. For $e \sim \text{Uniform}(-\Delta/2, \Delta/2)$, $\mathbb{E}[e^2] = \int_{-\Delta/2}^{\Delta/2} e^2 \cdot \frac{1}{\Delta}\, de = \frac{1}{\Delta}\left[\frac{e^3}{3}\right]_{-\Delta/2}^{\Delta/2} = \frac{1}{3\Delta} \cdot \frac{\Delta^3}{4} = \frac{\Delta^2}{12}$. □

TABLE 1
Quantization error: theory vs. simulation

Bits	Δ	Max err (emp.)	Max err (theory)
8	0.038276	0.019138	0.019138
4	0.612423	0.306211	0.306211

TABLE 2
Mean squared error: theory vs. simulation

Bits	MSE (emp.)	MSE (theory $= \Delta^2/12$)
8	0.00012216	0.00012209
4	0.03126167	0.03125514

1.2 Worked numerical verification

We quantize $2{,}000{,}000$ simulated weight values drawn from a standard normal distribution, at two bit depths:

The maximum-error bound is matched *exactly* (as it must be — Proposition 1 is a hard bound, not a statistical approximation), and the mean-squared-error prediction matches to four significant figures. Halving the bit depth from 8 to 4 increases Δ by a factor of 16.0 $(0.612423/0.038276)$, and MSE increases by very close to $16^2 = 256\times$ $(0.03126/0.000122 \approx 256.0)$, exactly as Eq. (3) predicts, since MSE scales with Δ^2.

1.3 Relation to the fixed-bit-budget principle

This confirms the distinction drawn in prior work between rescaling (lossless) and fixed-bit-budget quantization (lossy): the loss here comes entirely from the number of representable levels (2^b) being finite, not from the size of the range $[x_{\min}, x_{\max}]$ itself. Doubling the range while keeping b fixed doubles Δ and therefore quadruples MSE — the identical mechanism as an ADC with fixed bit depth losing resolution when asked to cover a larger physical range.

2 Embedding Search: Where Scaling Is Provably Lossless

Semantic search, recommendation systems, and retrieval-augmented generation typically rank items by cosine similarity between embedding vectors:

$$\cos(u, v) = \frac{u \cdot v}{\|u\| \, \|v\|}. \tag{3}$$

Proposition 3 (Scale invariance of cosine similarity)**.** *For any $u, v \in \mathbb{R}^d \setminus \{0\}$ and any scalars $s, t > 0$,*

$$\cos(su, tv) = \cos(u, v). \tag{4}$$

Proof. $\cos(su, tv) = \dfrac{(su) \cdot (tv)}{\|su\| \, \|tv\|} = \dfrac{st \, (u \cdot v)}{s\|u\| \cdot t\|v\|} = \dfrac{st}{st} \cdot \dfrac{u \cdot v}{\|u\|\|v\|} = \cos(u, v)$, since norms scale linearly with positive scalar multiplication: $\|su\| = s\|u\|$ for $s > 0$. □

2.1 Worked numerical verification

Two random 128-dimensional vectors give a baseline cosine similarity of 0.1159099926. Independently rescaling each vector by very different positive factors — $(2.0, 1.0)$, $(0.1, 5.0)$, and even the extreme case $(100.0, 0.01)$ — reproduces 0.1159099926 in every case, to all ten displayed decimal digits. This confirms Proposition 3 holds exactly, not approximately, even under a $10{,}000\times$ disparity in relative scale between the two vectors.

2.2 Practical consequence

This is why embedding-based search is robust to differences in vector magnitude arising from unrelated causes (e.g., document length, unnormalized model outputs, or differing encoder versions) as long as cosine similarity — rather than raw dot product or Euclidean distance, neither of which is scale-invariant — is the ranking function. It is also why some retrieval systems explicitly normalize embeddings to unit length: doing so makes cosine similarity and dot product coincide, without changing the ranking cosine similarity alone would already have produced.

3 Conclusion

The two mechanisms examined here sit on opposite sides of the same boundary established in earlier work:

- **Quantization** is lossy because it combines rescaling with rounding to a fixed, finite set of levels (2^b); the loss is exactly characterized by Eqs. (1)–(3) and confirmed numerically to four significant figures.
- **Cosine-similarity embedding search** is lossless under positive rescaling because it is a pure ratio, structurally identical to the ratio-preservation property proved for linear rescaling in the original correction — confirmed here to ten decimal digits even under extreme, mismatched scaling.

As in the rest of this series, the underlying question is always the same: is a given transformation a bijection preserving relative structure (lossless), or does it additionally discretize or truncate onto a fixed, finite budget (lossy)? Every result in this series reduces to correctly answering that one question for the specific transformation at hand.

References

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3) B. Jacob et al., “Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference,” CVPR, 2018.